MULTIMODAL SPATIAL BIOLOGY SERIES | VIDEO 2B

How we register H&E, COMET and Xenium images

A plain-language guide to the approach, the evidence behind it, and its limits. Companion to the video and to the free app at github.com/bnsreenu/spatial-bio-registration (version 0.2.0, MIT license).

1. The problem

A tissue core can be imaged three ways: an H&E stain (a color photo of the cells), COMET (16 protein channels, including DAPI, which lights up nuclei), and Xenium (the positions of individual transcripts, plus the outlines of nuclei). They are all pictures of the same piece of tissue, but each lives in its own coordinate system. Pixel sizes differ (about 0.203, 0.28 and 0.2125 micrometers per pixel), and each image is rotated and shifted a little differently.

Before you can ask whether a cell that looks like a tumor cell in H&E also expresses a certain protein and transcript, you need a map from one image to another. Building that map is called registration.

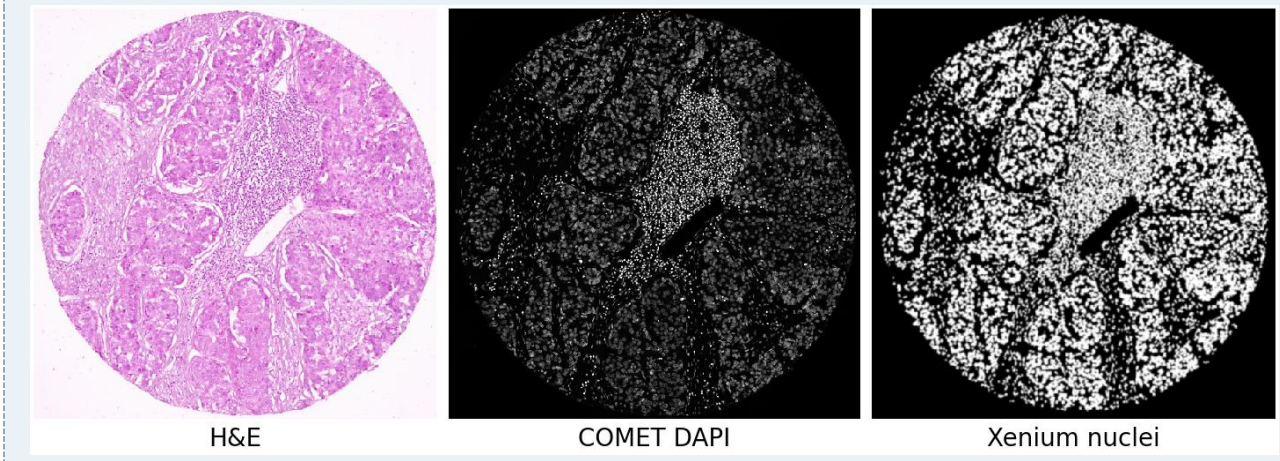


Figure 1. The same tissue in three modalities.

2. The big idea: nuclei are shared landmarks

Every modality can show nuclei. H&E shows them as blue-purple blobs, COMET shows them in the DAPI channel, and Xenium gives us their outlines. If we reduce each image to a picture of nuclei, the three pictures look alike, and we can find the same nuclei in each.

The output is a transform, not an image. We never resample or rewrite your images. The app saves a small matrix (plus matching keypoints and a parameters file) that tells any downstream tool how to move coordinates from one image to another. Your originals stay untouched.

What a transform contains

We use a similarity transform. It allows exactly four things: rotate the image, scale it uniformly, and shift it in x and y. Written as a 3 by 3 matrix, a point (x, y) in the moving image lands at (x′, y′) in the fixed frame:

$$[x' \;\; y' \;\; 1]^{T} = \begin{bmatrix} s\cdot\cos\theta & -s\cdot\sin\theta & tx \\ s\cdot\sin\theta & s\cdot\cos\theta & ty \\ 0 & 0 & 1 \end{bmatrix} [x \;\; y \;\; 1]^{T}$$

Here s is the scale, θ the rotation, and tx, ty the shift. The bottom row (0, 0, 1) is bookkeeping that lets one matrix carry all four numbers.

A worked example

From the core01 COMET matrix in the video 2 notebook, the matrix entries are a = 1.31386 and b = −0.01720, so the scale is √(a² + b²) = 1.314 and the rotation is atan2(b, a) = −0.75 degrees. The COMET pixel at (1000, 2000) lands at (9338.8, 21080.6) pixels in the Xenium DAPI frame. Multiplying by 0.2125 micrometers per pixel gives (1984.5, 4479.6) micrometers.

3. How the app finds the transform

Step 1. Make the modalities look alike

All three modalities are converted to a grayscale picture of nuclei at 1 micrometer per pixel. For H&E we separate the hematoxylin stain with color deconvolution. For COMET we use the DAPI channel. This dataset has no Xenium image, so we draw the nuclei from the nucleus outlines (nucleus_boundaries.parquet). At 1 micrometer per pixel, one pixel equals one micrometer, which keeps every later threshold easy to read.

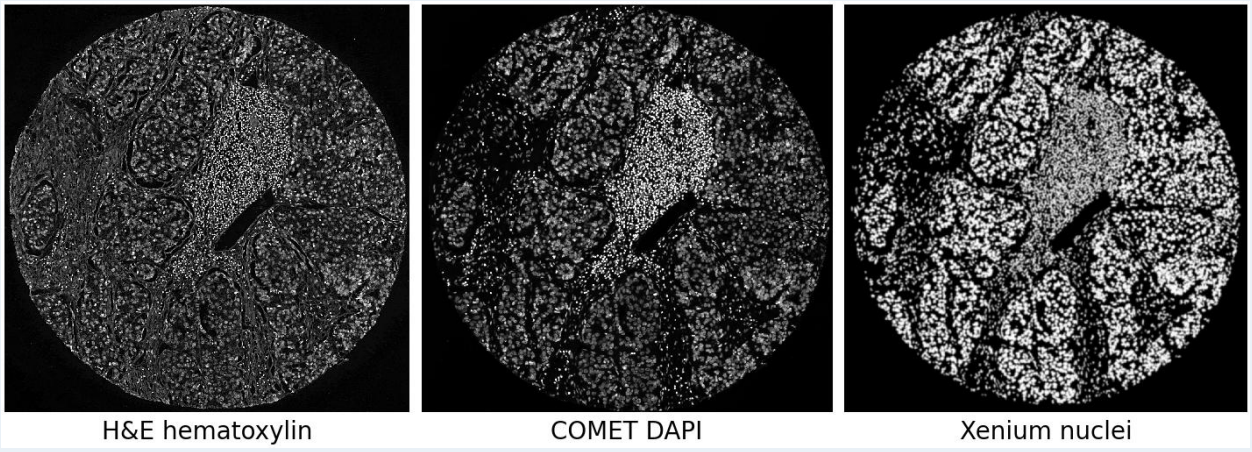


Figure 2. The three working images that the matching step sees.

Step 2. Find matching points

A feature detector called AKAZE picks out distinctive spots in each image and gives each a fingerprint. We pair spots whose fingerprints are alike. To avoid ambiguous pairs, we apply the ratio test: a pair is kept only if its best match is clearly better than the second best (the ratio must be under 0.75). A second detector, SIFT, repeats the process as an independent check.

Step 3. Keep only the matches that agree

Even after the ratio test, some pairs are wrong. A method called RANSAC handles this. It repeatedly proposes a transform from a few random pairs, counts how many other pairs agree with it (within 3 micrometers), and keeps the transform that the most pairs support. The agreeing pairs are called inliers. For example, matching the core01 H&E image to COMET gave 3029 pairs, and 2945 of them (97%) were inliers.

We also test the mirror-image possibility. A flipped image is accepted only if it fits better than the unflipped one.

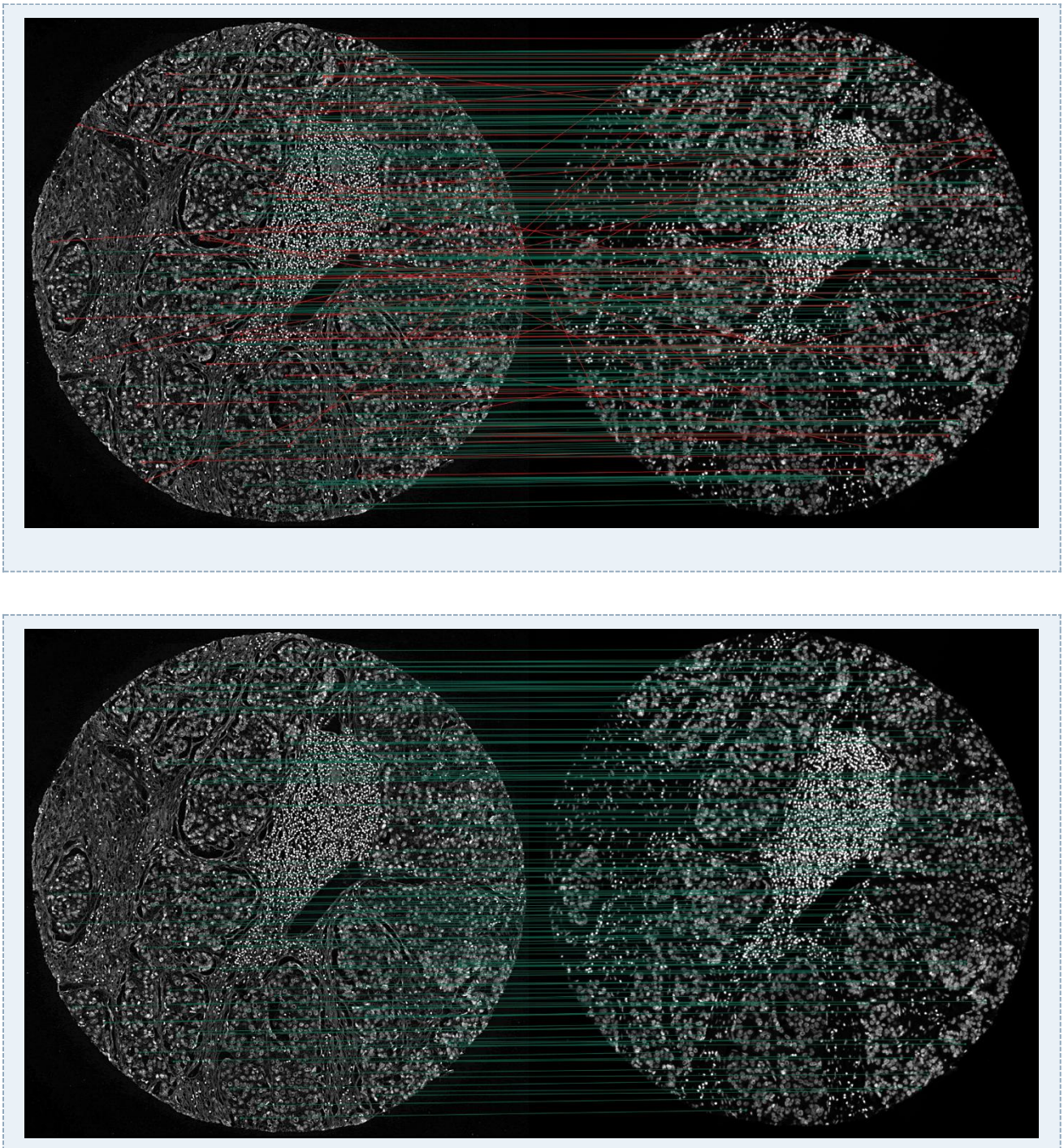


Figure 3. H&E to COMET matches in core01, before and after RANSAC.

Step 4. Chain the steps and choose a reference

H&E and Xenium nuclei look quite different, so we register in two steps: H&E to COMET, then COMET to Xenium. COMET DAPI is the bridge. The two matrices are multiplied to map H&E directly into the Xenium frame.

You can choose any image as the reference. Because a similarity matrix can be inverted and matrices can be multiplied, the app computes the other directions from the two it already has, without registering again. The default reference is Xenium, then COMET, then H&E.

4. How to trust a result

Every registration carries a confidence label with a plain-language reason.

- **High:** at least 100 inliers and at least 70% of pairs agree.
- **Low:** fewer than 30 inliers, or fewer than 40% of pairs agree.
- **Medium:** everything in between. Look at the overlay before relying on it.

The app warns you when SIFT disagrees with AKAZE by more than 1 micrometer. It also checks for tissue deformation by fitting a more flexible model (a full affine, which also allows stretch and shear) to the same matches. If the flexible model fits clearly better, or the leftover errors have a smooth spatial pattern, the app flags possible deformation and caps confidence at medium. An optional checkbox lets you export the affine matrix instead. Whether Xenium Explorer accepts a non-similarity matrix has not been verified.

5. How we tested it

Agreement with the dataset alignments

The dataset ships its own alignments, made from hand-picked landmarks. We compared our automatic result with them on a grid of points across the tissue in all four cores.

Core (mean / worst, µm)	COMET to fixed	H&E to fixed	H&E to COMET
core01 HCC	0.55 / 0.74	0.32 / 0.53	0.28 / 0.32
core02 non-tumour	0.48 / 0.58	0.40 / 0.68	0.23 / 0.42
core03 tonsil	0.35 / 0.64	0.48 / 0.87	0.46 / 0.91
core04 HCA	0.19 / 0.27	0.56 / 0.84	0.41 / 0.67

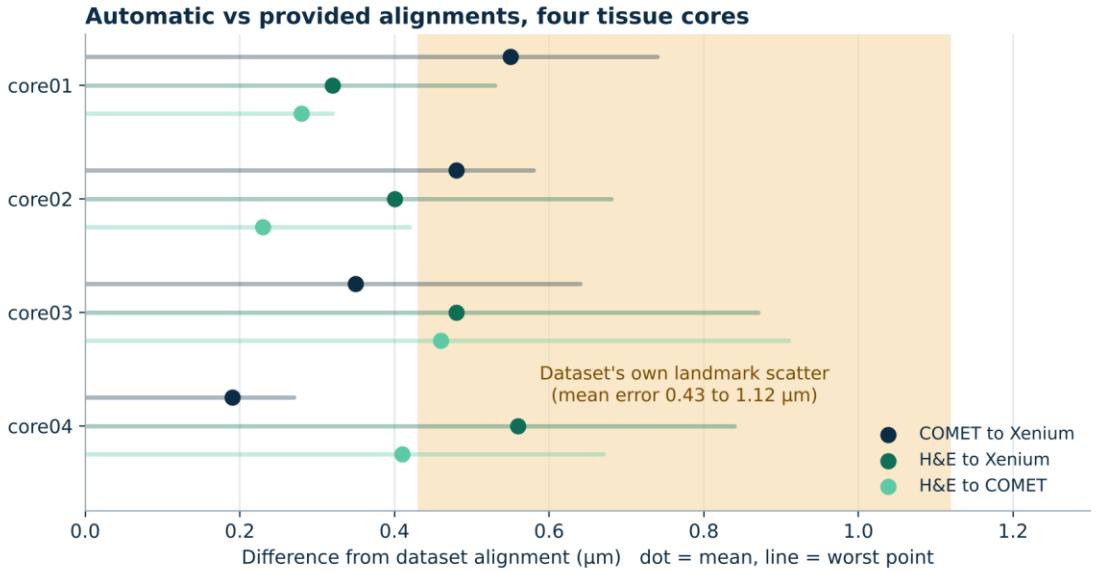


Figure 4. Differences are smaller than the dataset's own landmark scatter.

The provided alignments leave a mean error of 0.43 to 1.12 micrometers at their own landmarks, with worst cases of 1.2 to 6.7 micrometers. Our differences (mean 0.19 to 0.56, worst 0.91) are within that scatter. So we claim agreement to within about 1 micrometer, not that either alignment is correct to better than that. An independent nuclei-matching check scored the automatic result slightly better in 6 of 8 core-step cases and slightly worse in 2, always by small amounts under a detection noise floor of about 1 micrometer. Repeating a run gives identical matrices.

Stress tests with known answers

We took real tissue images from two cores, deliberately moved one image by a known amount, and checked whether the app recovered it (180 runs, both steps). Everything a similarity can represent was recovered: shifts up to 500 micrometers, rotations from 1 to 270 degrees, mirror flips, scale changes of ±10%, and a declared pixel size that was 2% or 5% too large. All 88 of these runs were accurate (mean error 0.04 to 0.47 micrometers, worst single point 0.87).

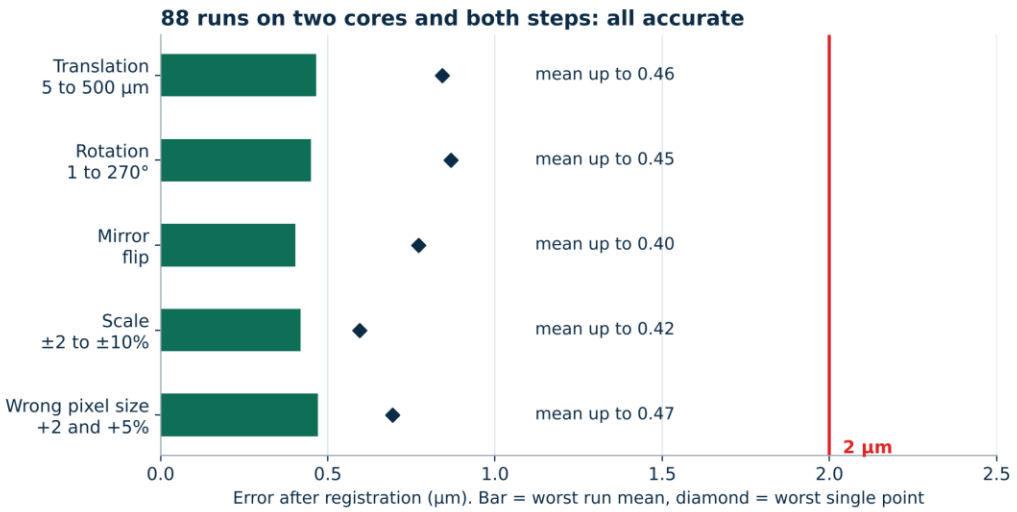


Figure 5. Rigid motions: every run stays well below the 2 micrometer line.

Where it breaks, and what fixes it

A similarity cannot represent stretch, shear or non-linear warping, so error grows with them. Stretch of 1% already costs about 2 micrometers, and 5% costs about 13. One degree of shear costs about 4 micrometers. A 10 micrometer peak elastic warp costs about 3. The affine option removes about 97% of the error from stretch and shear, but does not help with elastic warping, which the app flags rather than corrects. Error from unmodeled stretch should also grow with the size of the field of view. The cores here are about 1.2 to 1.9 millimeters across; we did not measure other sizes.

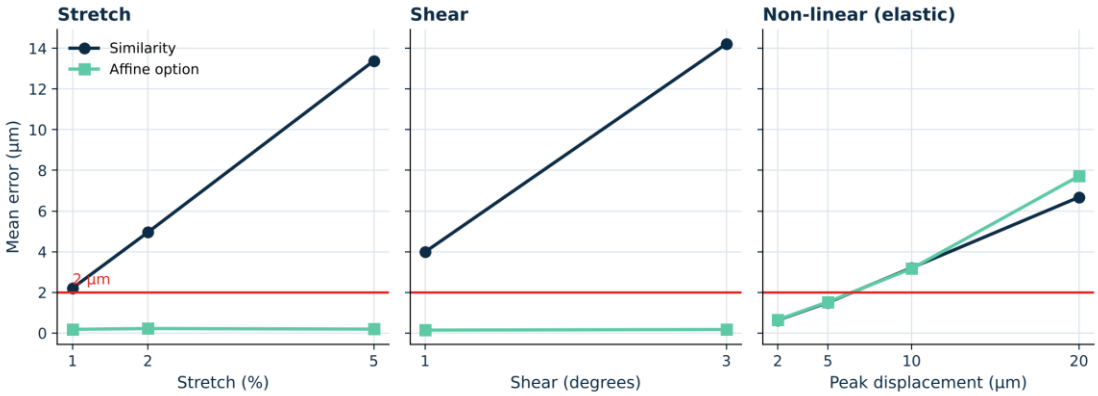


Figure 6. Synthetic deformation. Red line: 2 micrometers.

Does the confidence label tell the truth?

Across the 180 stress runs, none of the 36 low-confidence results were accurate, and 100 of 101 high-confidence results were. Medium was mixed (18 accurate, 21 not). The deformation check then fixed the weak spot: on four cores and 112 runs it flagged 32 of 32 stretch and shear runs and 31 of 32 elastic runs, found none on the eight real steps, and with the confidence cap in place all 46 high-confidence runs were accurate. It also raised 3 false alarms in 51 undeformed runs and missed 1 of 61 deformed runs. All deformation here is synthetic, and the thresholds were set on two cores and checked on four.

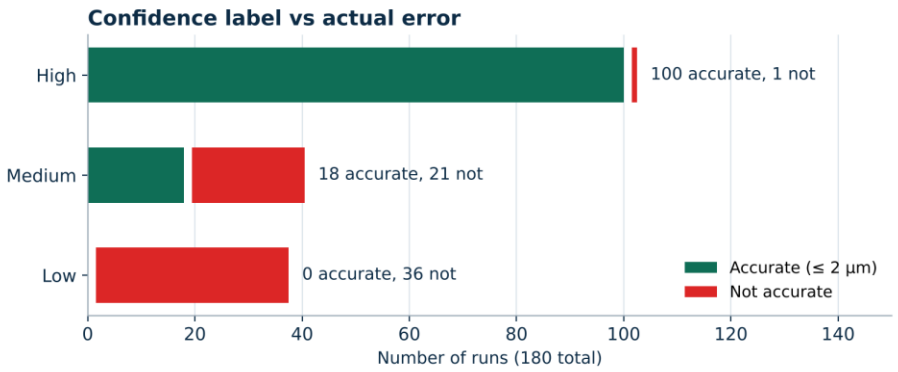


Figure 7. Confidence label against measured error (before the deformation cap).

6. What it can and cannot do

Tested

- Four cores from one dataset (UnumLocalia), real OME-Zarr data, same-section H&E, COMET and Xenium.
- Rotation, flip, scale and shift, with known answers.
- Windows with Python 3.10.

Not yet tested

- Serial sections (neighboring slices, not the same section).
- Real SVS or NDPI slides and real Xenium morphology images (only generated files were used).
- Xenium Explorer opening an affine matrix.
- Real strong tissue deformation. None appeared in these four cores.
- Manual landmark correction. This version does not include it.

A note on pixel sizes

Metadata pixel sizes can disagree with the images. The H&E pixel size implied by both the dataset alignments and our result is about 0.2000 micrometers, roughly 1.4% below the stated 0.203125. COMET is about 0.3% below its stated 0.28. Because the app fits the scale from the image content, it is not thrown off by this, but do not assume the metadata is exact.

7. Using the app

Put each sample in a folder with he, comet and xenium subfolders. Run python -m app, add the folder, click Run auto-registration, then review the overlay and confidence and save. Results go to a results folder per sample: a matrix file per modality, a keypoints file in Xenium Explorer format, a params.json, and a batch_summary.csv when you run several samples.

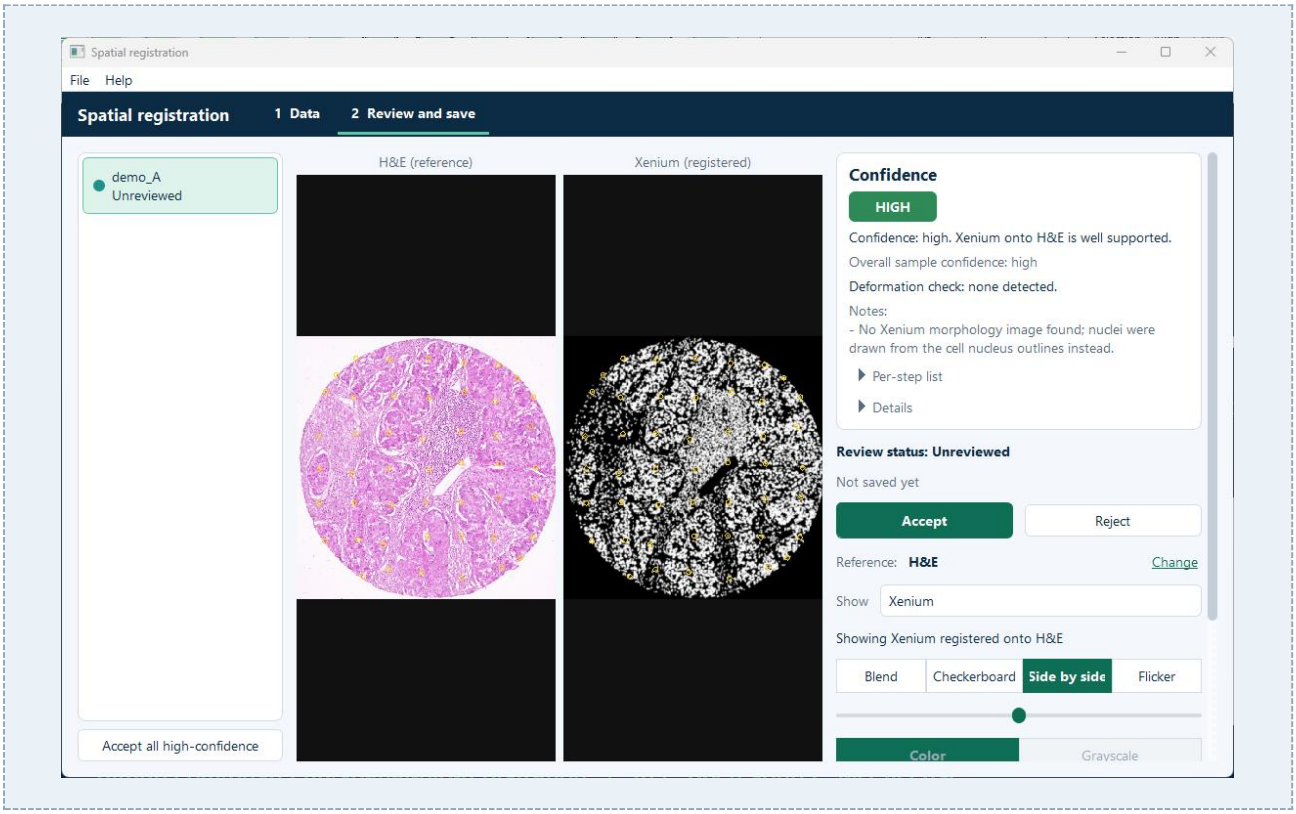


Figure 8. The review screen.

Glossary

Term	Meaning
Registration	Finding the map between two images of the same tissue
Similarity transform	Rotation, uniform scale and shift, nothing else
Affine transform	A similarity plus stretch and shear
Keypoint	A distinctive spot a detector can find again in another image
Inlier	A match that agrees with the final transform
RANSAC	Repeated random guess-and-check that ignores wrong matches
AKAZE, SIFT	Two keypoint detectors; SIFT is the second opinion
Fixed frame	The coordinate system everything is mapped into

Code: github.com/bnsreenu/spatial-bio-registration. Alpha release; please read the limits in the README before relying on results.